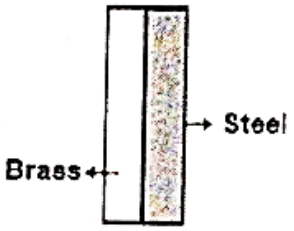
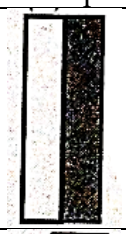
Question	When a uniform rod is heated, which of its following properties will increase as a result of it?
Type	multiple choice
Option	Mass
Option	Weight
Option	center of mass
Option	moment of inertia
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A copper rod of 88 cm and an aluminum rod of an unknown length have an equal increase in their lengths independent of an increase in temperature. The length of the aluminum rod is: $\left(\alpha_{Cu} = 1.7\times10^{-5}K^{-1}\ and\ \alpha_{Al} = 2.2\times10^{-5}K^{-1}\right)$
Type	multiple choice
Option	68 cm
Option	6.8 cm
Option	113.9 cm
Option	88 cm
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The value of the coefficient of volume expansion of glycerin is $5\times10^{-4}K^{-1}$. The fractional change in the density of glycerin for a temperature increase of 40°C will be:
Type	multiple choice
Option	0.015
Option	0.020
Option	0.025
Option	0.010
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The coefficients of linear expansion of brass and steel rods area α_1 and α_2, lengths of brass and steel rods are l_1 and l_2 respectively. If $(l_2 - l_1)$ is maintained the same at all temperatures, which one of the following relations hold good?
Type	multiple choice
Option	$\alpha_1 l_2^2 = a_2 l_1^2$
Option	$\alpha_1 2l_2 = \alpha_2^2 l_1$
Option	$\alpha_1 l_1 = \alpha_2 l_2$
Option	$\alpha_1 l_2 = \alpha_2 l_1$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If two rods of length L and 2L having coefficients of linear expansion a and 2a respectively are connected so that their total length becomes 3L, the average coefficient of linear expansion of the composition of rods equals:

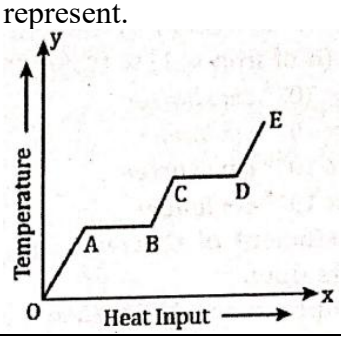
Type	multiple choice
Option	$\frac{3}{2}\alpha$
Option	$\frac{5}{2}\alpha$
Option	$\frac{5}{3}\alpha$
Option	None of these
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A pendulum clock runs faster by 5 seconds per day at 20°C and goes slow by 10 second per day at 35°C. It shows the correct time at a temperature of
Type	multiple choice
Option	$27.5^0 C$
Option	$25^0 C$
Option	$30^0 C$
Option	$33^0 C$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A bimetallic strip is shown in the figure. On cooling, which shape will it attain if $\alpha_{brass} > \alpha_{steel}$: 
Type	multiple choice
Option	
Option	
Option	

Option	
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If α, β and γ are coefficient of linear, area and volume expansion respectively, then
Type	multiple choice
Option	$\gamma = 3\alpha$
Option	$\alpha = 3\gamma$
Option	$\gamma = 3\beta$
Option	$\beta = 3\alpha$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A solid ball of metal has a spherical cavity inside it. The ball is heated. The 10. volume of cavity will
Type	multiple choice
Option	Increase
Option	Decrease
Option	remains unchanged
Option	have its shape changed
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A bar iron is 10 cm at 20°C. At 19°C it will be $(\alpha$ of iron $= 11\times10^{-6}/^{0}C$
Type	multiple choice
Option	$11\times10^{-5} cm\ shorter$
Option	$11\times10^{-6} cm\ longer$
Option	$11\times10^{-6} cm\ shorter$
Option	$11\times10^{-5} cm\ longer$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The coefficient of thermal conductivity depends upon
Type	multiple choice
Option	Temperature difference of two surfaces.
Option	Area of the plate
Option	Thickness of the plate
Option	Material of the plate
Answer	4
Solution	
Positive Marks	4

Negative Marks	1
Question	When two ends of a rod wrapped with cotton are maintained at different temperatures and after some time, every point of the rod attains a constant temperature, then;
Type	multiple choice
Option	Conduction of heat at different points of the rod stops because the temperature is not increasing
Option	The rod is a bad conductor of heat
Option	Heat is being radiated from each point of the rod
Option	Each point of the rod is giving heat to its neighbor at the same rate at which it is receiving heat
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is
Type	multiple choice
Option	$\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$
Option	$\frac{\sigma_1\sigma_2}{2\sigma_1\sigma_2}$
Option	$\frac{\sigma_1+\sigma_2}{\sigma_1\sigma_2}$
Option	$\frac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The ratio of thermal conductivity of two rods of different material is 5:4. The two rods of same area of cross-section and same thermal resistance will have the lengths in the ratio
Type	multiple choice
Option	$4:5$
Option	$9:1$
Option	$1:9$
Option	$5:4$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A slab consists of two parallel layers of copper and brass of the same thickness and having thermal conductivities in the ratio 1:4. If the free face of brass is at 100°C and that copper at 0°C, the temperature of interface is
Type	multiple choice
Option	$80^0 C$
Option	$20^0 C$
Option	$60^0 C$

Option	$40^0 C$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two thin blankets keep more hotness than one blanket of thickness equal to these two. The reason is
Type	multiple choice
Option	Their surface area increases
Option	A layer of air is formed between these two blankets, which is bad conductor
Option	These have more wool
Option	They absorb more heat from outside
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The temperature of the hot and cold ends of a 20 cm long rod in a thermal steady state is at 100°C and 20°C respectively. The temperature at the center of the rod will be
Type	multiple choice
Option	$50^0 C$
Option	$60^0 C$
Option	$40^0 C$
Option	$30^0 C$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Consider a compound slab consisting of two different materials having an equal thickness and thermal conductivities K and 2K respectively. The equivalent thermal conductivity of the slab if the materials are joined by ends is:
Type	multiple choice
Option	$\sqrt{2K}$
Option	$3K$
Option	$\frac{4}{3}K$
Option	$\frac{2}{3}K$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of length 1m having cross sectional area $0.75m^2$ has heat flow through it at the rate of 6000 joule/sec. Then find the temperature difference if $K = 200\ Jm^{-1}K^{-1}$
Type	multiple choice
Option	$20^0 C$
Option	$40^0 C$
Option	$80^0 C$

Option	$100^0 C$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Under steady state, the temperature of a body
Type	multiple choice
Option	Increases with time
Option	Decreases with time
Option	Does not change with time and is same at all the points of the body
Option	Does not change with time but is different at different points of the body
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The thermal capacity of any body is.
Type	multiple choice
Option	A measure of its capacity to absorb heat
Option	A measure of its capacity to provide heat
Option	The quantity of heat required to raise its temperature by a unit degree
Option	The quantity of heat required to raise the temperature of a unit mass of the body by a unit degree
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The amount of heat required to convert 1 g of ice at 0°C into steam at 100°C, is $\left(L_{steam} = 536\ cal/g\right)$
Type	multiple choice
Option	716 cal
Option	500 cal
Option	180 cal
Option	100 cal
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Work done in converting one gram of ice at $-10^0 C$ into steam at $100^0 C$ is $\left(L_{steam} = 540 cal/g\right)$
Type	multiple choice
Option	3045 J
Option	6056 J
Option	721 J
Option	616 J
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A water fall is 84 metres high. If half of the potential energy of the falling water gets converted to heat, the rise in temperature of water will be.
Type	multiple choice
Option	$0.098^0 C$

Option	$0.98^0 C$
Option	$9.8^0 C$
Option	$0.0098^0 C$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A bullet moving with velocity v collides against wall. consequently, half of its kinetic energy is converted into heat. If the whole heat is acquired by the bullet, the rise in temperature will be.
Type	multiple choice
Option	$v^2 / 4S$
Option	$4v^2 / 2S$
Option	$v^2 / 2S$
Option	v^2 / S
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	$2kg$ ice at $-20^0 C$ is mixed with $5kg$ water at $20^0 C$. Then final amount of water in the mixture would be. Given specific heat of ice $= 0.5 cal / g^0 C$, Specific heat of water $= 1 cal / g^0 C$, Latent heat of fusion for ice $= 80 cal / g$.
Type	multiple choice
Option	6 kg
Option	5 kg
Option	4 kg
Option	2 kg
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A solid material is supplied with heat at a constant rate. The temperature of material is changing with heat input as shown in the figure. What does slope DE represent. 
Type	multiple choice
Option	Latent heat of liquid
Option	Latent heat of vapour
Option	Heat capacity of vapour
Option	Inverse of heat capacity of vapour
Answer	4
Solution	
Positive Marks	4
Negative Marks	1

Question	A block of mass 2.5 kg is heated to temperature of 500°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx.). Specific heat for the body $= 0.1 cal / g^0 C.$
Type	multiple choice
Option	1 kg
Option	1.56 kg
Option	2 kg
Option	2.54 kg
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	10 g of ice at 0°C is kept in a calorimeter of water equivalent 10g. How much heat should be supplied to the apparatus to evaporate the water thus formed? (Neglect loss of heat)
Type	multiple choice
Option	6200 cal
Option	7200 cal
Option	13600 cal
Option	8200 cal
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	2 litre water at 27°C is heated by a 1 kW heater in an open container. On an average heat is lost to surroundings at the rate 160J/s. The time required for the temperature to reach 77°C is?
Type	multiple choice
Option	8 min 20 sec
Option	10 min
Option	7 min
Option	14 min
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If wavelengths of maximum intensity of radiations emitted by sun and the moon are $0.5 \times 10^{-6} m$ and $10^{-4} m$ respectively, the ratio of their temperatures is.
Type	multiple choice
Option	1/100
Option	1/200
Option	100
Option	200
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Solar radiation emitted by sun resembles that emitted by a black body at a temperature of 6000 K. Maximum intensity is emitted at a wavelength of about 4800Å. If the sun were to cool down from 6000 K to 3000 K, then the peak intensity would occur at a wavelength.
Type	multiple choice
Option	4800 Å

Option	9600 Å
Option	7200 Å
Option	6400 Å
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A metal ball of surface area 200 cm² and temperature 527°C is surrounded by a vessel at 27°C. If the emissivity of the metal is 0.4, then the rate of loss of heat from the ball is $\left(\sigma = 5.67 \times 10^{-8} J / m^2 - s - k^4\right)$
Type	multiple choice
Option	108 joules approx.
Option	168 joules approx.
Option	182 joules approx.
Option	192 joules approx.
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The absorptive power of a perfectly black body is.
Type	multiple choice
Option	Zero
Option	Infinity
Option	1.5
Option	1.0
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If temperature of a black increases from 7°C to 287°C then the rate of energy radiation increases by.
Type	multiple choice
Option	$\left(\frac{287}{7}\right)^4$
Option	16
Option	4
Option	2
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Star A has radius r surface temperature T while star B has radius 4r and surface temperature T/2. The ratio of the power of two starts, A: B is
Type	multiple choice
Option	16 : 1
Option	1 : 16
Option	1 : 1
Option	1 : 4
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	Radius of two spheres of same material are 1 and 8 m respectively and their temperature are 4×10^3 and $2\times10^3\,K$ respectively. Then ratio of emitted energy of sphered per sec. will be.
Type	multiple choice
Option	$1:2$
Option	$2:1$
Option	$1:4$
Option	$4:1$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If a metallic sphere gets cooled from 62^0C to 50^0C in 10 minutes and in the next 10 minutes gets cooled to $42^0C,$ then the temperature of the surrounding is:
Type	multiple choice
Option	30^0C
Option	36^0C
Option	26^0C
Option	20^0C
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A body cools from 50.0°C to 49.9°C in 5 s. How long will it take to cool from 40.0°C to 39.9°C? Assume the temperature of surrounding to be 30.0°C and Newton's law of cooling to be valid.
Type	multiple choice
Option	$2.5\,s$
Option	$10\,s$
Option	$20\,s$
Option	$5\,s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Four identical calorimeters painted in different colours, are heated to same temperature and then allowed to cool in vacuum. Which will cool fastest?
Type	multiple choice
Option	One which is painted bright
Option	One which is painted thick white
Option	One which is painted thick black
Option	One which is painted bright white
Answer	3
Solution	
Positive Marks	4
Negative Marks	1